Scattering Amplitudes: From QFT to Modern Methods

Ernest Yeung

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Part I

Quantum Field Theory (QFT Foundations)

Chapter 1

Quantum Field Theory (QFT) Foundations

1.1 LSZ Reduction Formula

1.2 Path Integrals

1.3 Feynman Rules for Gauge Theories

1.4 Non-Abelian Gauge Theory

Chapter 2

Helicity States and Weyl Spinors

This chapter builds the spinor algebra that underlies everything in Parts II and III. The physical target is simple: massless particles carry definite *helicity* $h = \pm\frac{1}{2}$ (or ± 1 for gluons), and the two-component Weyl spinors λ_α and $\tilde{\lambda}_{\dot{\alpha}}$ are precisely the objects that represent those helicity eigenstates. We establish this connection carefully here so that Chapter 3 can use the spinor-bracket formalism without re-deriving its foundations.

2.1 Lorentz Group Representations

The Lorentz algebra $\mathfrak{so}(1,3)$ is (complexified) isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ via the non-Hermitian combinations

$$N_i \equiv \tfrac{1}{2}(J_i - iK_i), \qquad N_i^\dagger \equiv \tfrac{1}{2}(J_i + iK_i), \tag{2.1}$$

where J_i are rotation generators and K_i are boost generators. They satisfy $[N_i, N_j] = i\varepsilon_{ijk}N_k$, $[N_i^\dagger, N_j^\dagger] = i\varepsilon_{ijk}N_k^\dagger$, $[N_i, N_j^\dagger] = 0$, so irreducible representations are labelled by two half-integers $(n,\, n')$.

(n, n')	Srednicki	Dim	Field	Index type
$(0,0)$	$(1,1)$	1	scalar ϕ	none
$(\frac{1}{2},0)$	$(2,1)$	2	left-handed Weyl ψ_α	undotted
$(0,\frac{1}{2})$	$(1,2)$	2	right-handed Weyl $\tilde{\psi}_{\dot{\alpha}}$	dotted
$(\frac{1}{2},\frac{1}{2})$	$(2,2)$	4	vector A^μ	spacetime

The key point for helicity: under a rotation by angle θ about the momentum axis (the little group of a massless particle), a $(\frac{1}{2},0)$ field picks up phase $e^{-i\theta/2}$ (helicity $-\frac{1}{2}$) and a $(0,\frac{1}{2})$ field picks up $e^{+i\theta/2}$ (helicity $+\frac{1}{2}$).

2.2 Left-Handed Weyl Spinors ψ_α

A left-handed Weyl field $\psi_\alpha(x)$ lives in the $(\frac{1}{2},0)$ representation. Under an infinitesimal Lorentz transformation $\delta\omega_{\mu\nu}$:

$$\delta\psi_\alpha = \tfrac{i}{2}\,\delta\omega_{\mu\nu}\,(S_{\mathrm{L}}^{\mu\nu})_\alpha{}^\beta\,\psi_\beta. \tag{2.2}$$

The six generators are explicitly (Srednicki eqs. 34.9–34.10):

$$(S_{\mathrm{L}}^{ij})_\alpha{}^\beta = \tfrac{1}{2}\varepsilon^{ijk}\sigma_k, \qquad (S_{\mathrm{L}}^{k0})_\alpha{}^\beta = \tfrac{i}{2}\sigma_k, \tag{2.3}$$

where σ_k are the Pauli matrices:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.4}$$

These satisfy the Lorentz algebra $[S_L^{\mu\nu}, S_L^{\rho\sigma}] = i(g^{\nu\rho} S_L^{\mu\sigma} - \cdots)$. The boost generators S_L^{k0} are anti-Hermitian (the Lorentz group is non-compact), reflecting that boosts are not unitary.

2.3 Right-Handed Weyl Spinors $\tilde{\psi}_{\dot{\alpha}}$

Hermitian conjugation swaps $N_i \leftrightarrow N_i^\dagger$, sending $(\frac{1}{2}, 0) \to (0, \frac{1}{2})$. So

$$(\psi_\alpha)^\dagger \equiv \tilde{\psi}_{\dot{\alpha}}(x) \tag{2.5}$$

is a *right-handed* Weyl field with a *dotted* index. Its generators are (Srednicki eq. 34.17):

$$\boxed{(S_{\mathrm{R}}^{\mu\nu})_{\dot{\alpha}}{}^{\dot{\beta}} = -[(S_{\mathrm{L}}^{\mu\nu})_\alpha{}^\beta]^*} \tag{2.6}$$

Concretely this reverses the sign of boosts (consistent with $L \leftrightarrow R$ under parity) and from eq. (35.22):

$$S_{\mathrm{R}}^{\mu\nu} = -\tfrac{i}{4}(\bar{\sigma}^\mu \sigma^\nu - \bar{\sigma}^\nu \sigma^\mu). \tag{2.7}$$

Convention note. We adopt Srednicki's metric $g^{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1)$ and $\varepsilon^{12} = \varepsilon_{21} = +1$ throughout. The amplitudes literature often uses the same conventions; we flag any deviation when it appears.

2.4 The ε Symbol: SL$(2, \mathbb{C})$ Metric

From $(\frac{1}{2}, 0) \otimes (\frac{1}{2}, 0) = (0, 0)_A \oplus (1, 0)_S$ there is a unique (up to scale) invariant antisymmetric symbol. With Srednicki's normalization (eq. 34.22 / 35.1):

$$\varepsilon^{12} = \varepsilon_{21} = +1, \qquad \varepsilon^{21} = \varepsilon_{12} = -1, \tag{2.8}$$

so in matrix form:

$$\varepsilon_{\alpha\beta} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \varepsilon^{\alpha\beta} = \begin{pmatrix} 0 & +1 \\ -1 & 0 \end{pmatrix}, \qquad \varepsilon_{\alpha\beta}\varepsilon^{\beta\gamma} = \delta_\alpha{}^\gamma. \tag{2.9}$$

The same structure holds for dotted indices.

2.4.1 Raising and lowering

$$\psi^\alpha = \varepsilon^{\alpha\beta}\psi_\beta, \qquad \psi_\alpha = \varepsilon_{\alpha\beta}\psi^\beta. \tag{2.10}$$

Sign trap (Srednicki eq. 34.27):

$$\psi^a \chi_a = \varepsilon^{ab}\psi_b\chi_a = -\varepsilon^{ba}\psi_b\chi_a = -\psi_b\chi^b. \tag{2.11}$$

Hence $\psi^\alpha\chi_\alpha = -\psi_\alpha\chi^\alpha$: *index position matters and carries a sign.*

2.5 Lorentz-Invariant Spinor Products

The fundamental Lorentz-invariant bilinears are:

$$\psi\chi \equiv \psi^\alpha \chi_\alpha = \varepsilon^{\alpha\beta}\psi_\alpha\chi_\beta, \tag{2.12}$$

$$\tilde{\psi}\tilde{\chi} \equiv \tilde{\psi}_{\dot\alpha}\tilde{\chi}^{\dot\alpha} = \varepsilon_{\dot\alpha\dot\beta}\tilde{\psi}^{\dot\alpha}\tilde{\chi}^{\dot\beta}. \tag{2.13}$$

For Grassmann-valued fields, $\psi\chi = +\chi\psi$ (two minus signs cancel; Srednicki eq. 35.25), and $(\psi\chi)^\dagger = \tilde{\chi}\tilde{\psi}$.

These are the precursors of the spinor brackets $\langle ij\rangle$ and $[ij]$ introduced in Chapter 3: when $\psi_\alpha = \lambda_{i\alpha}$ and $\chi_\alpha = \lambda_{j\alpha}$ are the spinors of massless momenta p_i, p_j, then $\langle ij\rangle = \varepsilon^{\alpha\beta}\lambda_{i\alpha}\lambda_{j\beta}$ and $[ij] = \varepsilon_{\dot\alpha\dot\beta}\tilde{\lambda}_i^{\dot\alpha}\tilde{\lambda}_j^{\dot\beta}$.

2.6 The σ^μ Symbol: Vector–Spinor Dictionary

A field in the $(\frac{1}{2}, \frac{1}{2})$ representation carries one undotted and one dotted index. The dictionary to 4-vectors is (Srednicki eq. 34.28–34.30):

$$A_{\alpha\dot\alpha} = \sigma^\mu_{\alpha\dot\alpha} A_\mu, \qquad \sigma^\mu_{\alpha\dot\alpha} = (I, \vec{\sigma}). \tag{2.14}$$

Explicitly:

$$\sigma^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \ \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \ \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \ \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.15}$$

Raising both spinor indices via the ε symbols gives (Srednicki eq. 35.19–35.20):

$$\bar{\sigma}^{\mu\,\dot\alpha\alpha} \equiv \varepsilon^{\alpha\beta}\varepsilon^{\dot\alpha\dot\beta}\,\sigma^\mu_{\beta\dot\beta} = (I, -\vec{\sigma}). \tag{2.16}$$

2.6.1 Key identities

$$\sigma^\mu_{\alpha\dot\alpha}\,\sigma_{\mu\beta\dot\beta} = -2\,\varepsilon_{\alpha\beta}\,\varepsilon_{\dot\alpha\dot\beta}, \tag{35.4}$$

$$\varepsilon^{\alpha\beta}\varepsilon^{\dot\alpha\dot\beta}\,\sigma^\mu_{\alpha\dot\alpha}\,\sigma^\nu_{\beta\dot\beta} = -2\,g^{\mu\nu}, \tag{35.5}$$

$$\mathrm{tr}(\sigma^\mu\bar{\sigma}^\nu) \equiv \sigma^\mu_{\alpha\dot\alpha}\,\bar{\sigma}^{\nu\,\dot\alpha\alpha} = 2\,g^{\mu\nu}, \tag{2.17}$$

$$\sigma^\mu\bar{\sigma}^\nu + \sigma^\nu\bar{\sigma}^\mu = -2\,g^{\mu\nu}\,I_2. \tag{35-Clifford}$$

The last line is the Weyl-spinor analogue of the Dirac Clifford algebra. The generators are its antisymmetric part:

$$S_{\mathrm{L}}^{\mu\nu} = \tfrac{i}{4}(\sigma^\mu\bar{\sigma}^\nu - \sigma^\nu\bar{\sigma}^\mu). \tag{35.21}$$

2.6.2 Momentum bi-spinor

For any 4-momentum p^μ, define

$$p_{\alpha\dot\alpha} \equiv \sigma^\mu_{\alpha\dot\alpha}\,p_\mu = \begin{pmatrix} p^0 + p^3 & p^1 + ip^2 \\ p^1 - ip^2 & p^0 - p^3 \end{pmatrix}. \tag{2.18}$$

Then $\det(p_{\alpha\dot\alpha}) = (p^0)^2 - |\mathbf{p}|^2 = p^2$, so $p^2 = 0$ iff the matrix has rank 1 — the factorization $p_{\alpha\dot\alpha} = \lambda_\alpha\tilde{\lambda}_{\dot\alpha}$ is equivalent to masslessness. This is the starting point of Chapter 3.

2.7 Helicity from the Little Group

For a massless particle with momentum p^μ, the little group (the subgroup of the Lorentz group that fixes p^μ) is $ISO(2)$, whose compact part is $U(1)$. Under the $U(1)$ rotation by angle θ, a field of helicity h transforms as $e^{-2ih\theta}$.

In terms of spinors: the little-group generator rescales

$$\lambda_\alpha \to e^{i\theta/2}\lambda_\alpha, \qquad \tilde{\lambda}_{\dot\alpha} \to e^{-i\theta/2}\tilde{\lambda}_{\dot\alpha}, \tag{2.19}$$

i.e. the continuous parameter $t = e^{i\theta/2}$ acts as

$$\lambda \to t\lambda, \qquad \tilde{\lambda} \to t^{-1}\tilde{\lambda}. \tag{2.20}$$

A state created by λ_α has helicity $-\frac{1}{2}$ (left-handed, positive-chirality), and one created by $\tilde{\lambda}_{\dot\alpha}$ has helicity $+\frac{1}{2}$ (right-handed, negative-chirality).

For a spin-1 gluon the helicity-± 1 polarization vectors scale as $\varepsilon^\pm_\mu \to t^{\mp 2}\varepsilon^\pm_\mu$ under (2.20); this determines the form of $\varepsilon^\pm$ in terms of spinors (see §3.5).

2.8 The Weyl Lagrangian

For a single left-handed Weyl field ψ_α, the Lorentz-invariant hermitian Lagrangian is (Srednicki eq. 36.2):

$$\boxed{\mathcal{L} = i\tilde{\psi}_{\dot\alpha}\,\bar{\sigma}^{\mu\,\dot\alpha\alpha}\,\partial_\mu\psi_\alpha - \tfrac{1}{2}m\,\varepsilon^{\alpha\beta}\psi_\alpha\psi_\beta - \tfrac{1}{2}m\,\varepsilon_{\dot\alpha\dot\beta}\tilde{\psi}^{\dot\alpha}\tilde{\psi}^{\dot\beta}.} \tag{2.21}$$

The kinetic term is real modulo a total divergence. The mass term is a *Majorana mass*: it breaks the $U(1)$ symmetry $\psi \to e^{i\alpha}\psi$ and is non-zero because ψ_α are Grassmann-valued.

2.8.1 Equations of motion

Varying with respect to $\tilde{\psi}^{\dot\alpha}$:

$$-i\,\bar{\sigma}^{\mu\,\dot\alpha\alpha}\,\partial_\mu\psi_\alpha + m\,\tilde{\psi}^{\dot\alpha} = 0. \tag{36.4}$$

For $m = 0$ this is the **Weyl equation** for a massless left-handed spinor. It is the field equation whose plane-wave solutions give the spinors $\lambda_\alpha(p)$ used in amplitude calculations.

2.8.2 Gamma matrices: Weyl representation

Assembling ψ_α and $\tilde{\psi}_{\dot\alpha}$ into a 4-component Majorana spinor $\Psi = (\psi_\alpha, \tilde{\psi}^{\dot\alpha})^T$, the equations of motion take the Dirac form $(-i\gamma^\mu\partial_\mu + m)\Psi = 0$ with (Srednicki eq. 36.7):

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu_{\alpha\dot\beta} \\ \bar{\sigma}^{\mu\,\dot\alpha\beta} & 0 \end{pmatrix}, \qquad \{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu}I_4. \tag{36.7, 36.9}$$

The chiral projectors are $P_L = \frac{1}{2}(I - \gamma^5) = \mathrm{diag}(I_2, 0)$ and $P_R = \frac{1}{2}(I + \gamma^5) = \mathrm{diag}(0, I_2)$ with $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \mathrm{diag}(-I_2, +I_2)$.

2.9 Majorana and Dirac Fermions

Property	Majorana	Dirac
Weyl fields	1 (ψ)	2 (χ, ξ)
$U(1)$ symmetry	No	Yes
Conserved charge	None	$Q = \int j^0 \, d^3x$
Self-conjugate ($\Psi^C = \Psi$)	Yes	No
Lagrangian	$\frac{i}{2}\bar{\Psi}\gamma^\mu\partial_\mu\Psi - \frac{m}{2}\bar{\Psi}\Psi$	$i\bar{\Psi}\gamma^\mu\partial_\mu\Psi - m\bar{\Psi}\Psi$
Real d.o.f.	2	4

In amplitude calculations with external gluons (massless spin-1), the internal spinors are always effectively massless Weyl spinors; the Majorana/Dirac distinction matters for external fermion lines.

2.10 Summary: Spinor Toolkit

Object	Expression / Rule
Left-handed field	ψ_α (undotted index, helicity $-\frac{1}{2}$)
Right-handed field	$\tilde{\psi}_{\dot{\alpha}} = (\psi_\alpha)^\dagger$ (dotted, $+\frac{1}{2}$)
Rotation generator	$(S_L^{ij})_\alpha{}^\beta = \frac{1}{2}\varepsilon^{ijk}\sigma_k$
Boost generator	$(S_L^{k0})_\alpha{}^\beta = \frac{i}{2}\sigma_k$
R-generators	$S_R^{\mu\nu} = -[S_L^{\mu\nu}]^*$
ε normalization	$\varepsilon^{12} = \varepsilon_{21} = +1$
Raise/lower	$\psi^\alpha = \varepsilon^{\alpha\beta}\psi_\beta, \quad \psi_\alpha = \varepsilon_{\alpha\beta}\psi^\beta$
Sign identity	$\psi^\alpha\chi_\alpha = -\psi_\alpha\chi^\alpha$
Angle bracket	$\psi\chi = \varepsilon^{\alpha\beta}\psi_\alpha\chi_\beta = +\chi\psi$
Square bracket	$\tilde{\psi}\tilde{\chi} = \varepsilon_{\dot{\alpha}\dot{\beta}}\tilde{\psi}^{\dot{\alpha}}\tilde{\chi}^{\dot{\beta}}$
σ^μ	$(I, \vec{\sigma}), \quad \bar{\sigma}^\mu = (I, -\vec{\sigma})$
Trace	$\mathrm{tr}(\sigma^\mu\bar{\sigma}^\nu) = 2g^{\mu\nu}$
Clifford	$\sigma^\mu\bar{\sigma}^\nu + \sigma^\nu\bar{\sigma}^\mu = -2g^{\mu\nu}I_2$
Momentum bi-spinor	$p_{\alpha\dot{\alpha}} = \sigma^\mu_{\alpha\dot{\alpha}}p_\mu, \quad \det = p^2$
Little group	$\lambda \to t\lambda$, $\tilde{\lambda} \to t^{-1}\tilde{\lambda}$
Weyl Lagrangian	$i\tilde{\psi}\bar{\sigma}^\mu\partial_\mu\psi - \frac{m}{2}\psi\psi - \frac{m}{2}\tilde{\psi}\tilde{\psi}$
Weyl equation ($m = 0$)	$i\bar{\sigma}^{\mu\,\dot{\alpha}\alpha}\partial_\mu\psi_\alpha = 0$

Next: Chapter 3 uses the momentum bi-spinor factorization $p_{\alpha\dot{\alpha}} = \lambda_\alpha\tilde{\lambda}_{\dot{\alpha}}$ (which follows directly from $p^2 = 0$ and the rank-1 condition above) to build the spinor-bracket formalism and polarization vectors for massless gauge bosons.

Part II

Spinor-Helicity Formalism

Chapter 3

Spinor-Helicity Formalism

3.1 The Factorization of Null Momenta

For massless particles, the momentum p^μ satisfies $p^2 = 0$. In the Van der Waerden notation, the momentum bi-spinor is defined via the Pauli matrices $\sigma^\mu = (\mathbf{1}, \vec{\sigma}_{\text{Pauli}})$:

$$p_{\alpha\dot{\alpha}} = p^\mu (\sigma_\mu)_{\alpha\dot{\alpha}}. \tag{3.1}$$

Explicitly in 2Œ2 matrix form:

$$p_{\alpha\dot{\alpha}} = \begin{pmatrix} p^0 + p^3 & p^1 + ip^2 \\ p^1 - ip^2 & p^0 - p^3 \end{pmatrix}. \tag{3.2}$$

For massless p ($p^2 = 0$), $\det(p_{\alpha\dot{\alpha}}) = (p^0)^2 - |\mathbf{p}|^2 = 0$, so this matrix has rank 1 and factorizes as:

$$p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}} \tag{3.3}$$

where λ_α and $\tilde{\lambda}_{\dot{\alpha}}$ are commuting (bosonic) Weyl spinors (Srednicki Sec. 20.3, 42.1). This factorization is unique up to the little group scaling:

$$\lambda_\alpha \to t\lambda_\alpha, \quad \tilde{\lambda}_{\dot{\alpha}} \to t^{-1}\tilde{\lambda}_{\dot{\alpha}}, \quad t \in \mathbb{C}^*. \tag{3.4}$$

For real momenta $t \in U(1)$; for complex momenta λ and $\tilde{\lambda}$ are independent. A left-handed (positive helicity $+\frac{1}{2}$) Weyl spinor scales as $\lambda \to t\lambda$; a right-handed one as $\tilde{\lambda} \to t^{-1}\tilde{\lambda}$.

3.2 Spinor Brackets

The fundamental Lorentz-invariant contractions are (Srednicki eq. 42.16):

$$\langle ij \rangle = \langle ij \rangle = \epsilon^{\alpha\beta} \lambda_{i\alpha} \lambda_{j\beta} = \lambda_i^0 \lambda_j^1 - \lambda_i^1 \lambda_j^0 \tag{3.5}$$

$$[ij] = [ij] = \epsilon^{\dot{\alpha}\dot{\beta}} \tilde{\lambda}_{i\dot{\alpha}} \tilde{\lambda}_{j\dot{\beta}} \tag{3.6}$$

with $\epsilon^{01} = -\epsilon^{10} = 1$ (Levi-Civita). These are antisymmetric: $\langle ij \rangle = -\langle ji \rangle$, $[ij] = -[ji]$.

The dot product of two massless momenta is encoded by:

$$2\, p_i \cdot p_j = \mathrm{tr}(\not{p}_i \not{p}_j) = \langle ij \rangle [ji]. \tag{3.7}$$

In particular $p_i^2 = 0$ implies $\langle ii \rangle = [ii] = 0$.

3.3 Affine (Inhomogeneous) Spinor Coordinates

For explicit calculations (as in arXiv:2602.12176v1, Eq. (2)), one fixes the little-group gauge and writes the spinors in affine coordinates on $\mathbb{CP}^1$:

$$|i\rangle \equiv \lambda_i = \begin{pmatrix} 1 \\ z_i \end{pmatrix}, \qquad |i] \equiv \tilde{\lambda}_i = \omega_i \begin{pmatrix} 1 \\ \tilde{z}_i \end{pmatrix}, \tag{3.8}$$

where $z_i, \tilde{z}_i \in \mathbb{C}$ are inhomogeneous (affine) coordinates on the Riemann sphere and $\omega_i \in \mathbb{C}^*$ carries the little-group weight (so $\omega_i \to t^{-1}\omega_i$ under $\lambda \to t\lambda$). In this gauge the spinor brackets read:

$$\langle ij\rangle = z_j - z_i, \tag{3.9}$$

$$[ij] = \omega_i\omega_j\,(\tilde{z}_j - \tilde{z}_i). \tag{3.10}$$

The dot product identity (3.7) then becomes $2p_i \cdot p_j = (z_j - z_i)\,\omega_i\omega_j(\tilde{z}_i - \tilde{z}_j)$. This coordinate patch is particularly convenient for the half-collinear kinematics relevant to single-minus amplitudes (§4.4).

3.4 Schouten Identity

For any three spinors λ_i, λ_j, λ_k:

$$\langle ij\rangle\lambda_{k\alpha} + \langle jk\rangle\lambda_{i\alpha} + \langle ki\rangle\lambda_{j\alpha} = 0 \tag{3.11}$$

Equivalently:

$$\langle ij\rangle\langle kl\rangle = \langle ik\rangle\langle jl\rangle + \langle il\rangle\langle kj\rangle \tag{3.12}$$

3.5 Polarization Vectors

For a gluon of momentum p and helicity $\pm$:

$$\varepsilon^+_\mu = \frac{\langle r|\gamma_\mu|p]}{\sqrt{2}\langle rp\rangle} \tag{3.13}$$

$$\varepsilon^-_\mu = \frac{\langle p|\gamma_\mu|r]}{\sqrt{2}[pr]} \tag{3.14}$$

where r is an arbitrary reference spinor.

3.6 Momentum Twistors

For planar amplitudes, momentum twistors $\mathcal{Z}_i^A = (\lambda_i^\alpha, \mu_i^{\dot\alpha})$ provide a dual description where:

$$x_i^{\alpha\dot\alpha} - x_{i+1}^{\alpha\dot\alpha} = p_i^{\alpha\dot\alpha} = \lambda_i^\alpha\tilde{\lambda}_i^{\dot\alpha} \tag{3.15}$$

The dual conformal symmetry becomes manifest in these variables.

Chapter 4

Maximally Helicity Violating (MHV) Amplitudes

4.1 Why "Maximally Helicity Violating"?

For a tree-level n-gluon amplitude $A_n(h_1, \ldots, h_n)$ with $h_i = \pm 1$:

- Let m_- denote the number of negative-helicity gluons.
- Naive Yang–Mills vertices with all same helicity ($+\cdots+$ or $-\cdots-$) are strongly suppressed — amplitudes with $m_- = 0$ or $m_- = n$ vanish at tree level (and at all loops).
- Amplitudes with $m_- = 1$ (single-minus) were long conjectured to vanish; see §4.4.
- In twistor space (Witten 2004), tree amplitudes are supported on curves of degree $d = m_- - 1 + \ell$, so at tree level $\ell = 0$ the non-zero sector requires $0 \le m_- \le 4$ generically.

MHV = maximally helicity violating corresponds to $m_- = 2$, i.e. $m_+ = n - 2$. This is the maximum number of "wrong-helicity" gluons that gives a generically non-zero amplitude, in the sense that every cyclic ordering is non-zero. The name reflects that having only 2 negative-helicity gluons out of n is the extreme violation of a naive helicity-conservation intuition.

4.2 The Parke-Taylor Formula

For n-gluon scattering with legs j and k carrying negative helicity, the tree-level color-ordered amplitude is:

$$\boxed{A_n^{\mathrm{MHV}}(\ldots, j^-, \ldots, k^-, \ldots) = i\frac{\langle jk\rangle^4}{\langle 12\rangle\langle 23\rangle\cdots\langle n1\rangle}} \tag{4.1}$$

(momentum-conserving delta function $\delta^4(P)$ implicit). For the canonical choice $(j, k) = (1, 2)$ this reads $i\,\langle 12\rangle^4/(\langle 12\rangle\langle 23\rangle\cdots\langle n1\rangle)$.

This compact formula replaces $O(n!)$ Feynman diagrams with a single expression.

Little group weight check. Under $\lambda_i \to t_i\lambda_i$, $\tilde{\lambda}_i \to t_i^{-1}\tilde{\lambda}_i$, a helicity-$h_i$ gluon requires the amplitude to scale as $t_i^{-2h_i}$.

- Negative-helicity legs j, k: numerator $\langle jk\rangle^4 \sim t_j^4 t_k^4$, denom contributes $t_j^{-1}t_k^{-1}$ (one factor each), net $t_j^4/t_j^{-1} = t_j^{4+1}$... more carefully: $\langle jk\rangle^4$ in numerator gives weight $+4$ for j and $+4$ for k; each $\langle\cdot j\cdot\rangle$ in denominator gives -1; there are exactly 2 such factors, so net weight for j is $4 - 2 = +2 = -2h_j$ with $h_j = -1$. ✓
- Positive-helicity legs: appear twice in the denominator (once as $\langle i{,}i{+}1\rangle$ and once as $\langle i{-}1{,}i\rangle$), giving weight $-2 = -2(+1)$. ✓

4.3 Helicity Configuration Table

Helicity Configuration	Tree Amplitude	Status
$(-,-,+,+,\ldots,+)$	$\neq 0$	MHV
$(-,+,+,+,\ldots,+)$	$\neq 0$	MHV (Parke-Taylor)
$(+,-,+,+,\ldots,+)$	$\neq 0$	MHV (cyclic)
$(-,-,-,+,\ldots,+)$	$\neq 0$	NMHV
$(-,+,+,+,+,+)$	$\neq 0$	Single-minus (2026 result)
$(+,+,+,+,+,+)$	$= 0$	Vanishing

4.4 Single-Minus Amplitudes Are Nonzero

Recent work (Guevara, Lupsasca, Skinner, Strominger, Weil, arXiv:2602.12176) proves that single-minus ($m_- = 1$) tree amplitudes are non-zero for all $n \geq 4$.

$$A_n(1^-, 2^+, \ldots, n^+) \neq 0 \quad \forall\, n \geq 4. \tag{4.2}$$

This breaks the longstanding conjecture that only $m_- \geq 2$ configurations were generically nonvanishing.

Half-collinear locus (arXiv:2602.12176 Eq. (13)). The single-minus amplitude is supported on the *half-collinear locus*:

$$\langle ij \rangle = 0 \quad \forall\, i, j \in \{1, \ldots, n\}. \tag{4.3}$$

This means all λ_i are proportional, i.e. $\lambda_i = c_i\, \lambda_1$ for some $c_i \in \mathbb{C}$, so all n momenta are collinear. The $\delta(\langle 1a \rangle)$ factors in the ansatz below enforce this support explicitly for $a = 2, \ldots, n$.

Explicit single-minus ansatz (arXiv:2602.12176 Eq. (15)). The ansatz is:

$$\begin{aligned} \mathcal{A}_n(1^-, 2^+, \ldots, n^+) = i^{2-n}\, \frac{\langle r1 \rangle^{n+1}}{\langle r2 \rangle \langle r3 \rangle \cdots \langle rn \rangle}\, A_{1\cdots n} \prod_{a=2}^{n} \delta(\langle 1a \rangle) \\ \times\ \delta^2\Big(\sum_{i=1}^{n} \langle ri \rangle\, \tilde{\lambda}_i\Big) \end{aligned} \tag{4.4}$$

where r is the reference spinor, $A_{1\cdots n}$ is a scalar prefactor, and $\delta^2(\cdots)$ is a 2-component ($\dot{\alpha}$) delta function imposing a residual momentum constraint. In affine coordinates (§3.3), $\langle 1a \rangle = z_a - z_1$, so the delta functions set $z_a = z_1$ for all a.

MHV comparison (arXiv:2602.12176 Eq. (11)). For comparison, the MHV amplitude in the same notation with negative-helicity legs r, s reads:

$$\mathcal{A}_n^{\mathrm{MHV}} = i\, \langle rs \rangle^4\, \mathrm{PT}_{\mathrm{cyc}}\, \delta^4\Big(\sum_{k=1}^{n} p_k\Big), \tag{4.5}$$

where $\mathrm{PT}_{\mathrm{cyc}} \equiv 1/(\langle 12 \rangle \langle 23 \rangle \cdots \langle n1 \rangle)$ is the cyclic Parke-Taylor denominator. The full 4-momentum-conserving delta function $\delta^4(\sum_k p_k)$ is displayed explicitly here (it was left implicit in (4.5)). The structural contrast with (4.4) is striking: MHV has a standard $\delta^4(P)$ on generic kinematics, while single-minus is localized on the half-collinear locus (4.3) via the $\prod_a \delta(\langle 1a \rangle)$ factors.

4.5 BCJ Relations

The Bern-Carrasco-Johansson (BCJ) relations express amplitude redundancies:

$$\sum_{i=2}^{n-1} \frac{s_{1i}}{\langle 1i\rangle[i1]} A_n(2,\ldots,i,1,i+1,\ldots,n) = 0 \tag{4.6}$$

These reduce the number of independent color-ordered amplitudes from $(n-3)!$ to $(n-3)!/2$.

4.6 Double Copy

Gauge theory amplitudes factorize into gravity amplitudes via:

$$M_n^{\text{gravity}} = \sum_{\text{duals}} \frac{A_n^{\text{gauge}} \tilde{A}_n^{\text{gauge}}}{\text{kinematic factors}} \tag{4.7}$$

This is the KLT (Kawai-Lewellen-Tye) relation.

Chapter 5

On-Shell Recursion (BCFW)

5.1 The BCFW Shift

Choose two legs i and j. Deform their spinors:

$$\lambda_i \to \lambda_i + z\lambda_j \tag{5.1}$$

$$\tilde{\lambda}_j \to \tilde{\lambda}_j - z\tilde{\lambda}_i \tag{5.2}$$

Under this shift:

$$p_i(z) + p_j(z) = p_i + p_j \quad \text{(momentum conserved)} \tag{5.3}$$

5.2 The Recursion Formula

If $A_n(z) \to 0$ as $z \to \infty$, then by Cauchy's theorem:

$$\boxed{A_n = \sum_{\text{channels } P} A_L^{\text{on-shell}} \frac{1}{P^2} A_R^{\text{on-shell}}} \tag{5.4}$$

where the sum runs over all factorization channels where P is on-shell ($P^2 = 0$).

5.3 Why It Works

Traditional Feynman diagrams explode combinatorially:

n gluons	Feynman diagrams
4	4
5	25
6	220
10	$\sim 10^7$

BCFW recursion reduces this to $O(n^3)$ complexity.

5.4 Application to Single-Minus

The arXiv:2602.12176 paper uses BCFW with:

- All-plus reference leg
- Single-minus shifted leg
- Induction on n

This proves the nonvanishing of single-minus amplitudes for all n.

5.5 Soft Limits

As $p_n \to 0$:

$$A_n \to \frac{\langle n-1, n\rangle}{\langle n, 1\rangle} A_{n-1} \tag{5.5}$$

This provides a consistency check on amplitude formulas.

Part III

Computational Tools

Chapter 6

Computational Tools

6.1 Recommended Tool Stack

Priority order for OpenClaw physics agent:

1. **lips** Spinor-helicity brackets, phase-space generation
2. **SymPy** Gamma matrix algebra, symbolic manipulation
3. **CADABRA2** Tensor calculus, field theory
4. **FORM** Large-scale polynomial algebra

6.2 lips

```
pip install lips
```

Features:

- Spinor brackets $\langle ij \rangle$, $[ij]$
- Massless phase-space over $\mathbb{C}$, $\mathbb{F}_p$, $\mathbb{Q}_p$
- Symbol alphabet generation
- 2-loop amplitudes via `antares-results`

6.3 SymPy

```
import sympy as sp
from sympy import symbols, simplify, latex

# Spinor bracket
def angle_bracket(i, j, lam):
    return lam[i][0]*lam[j][1] - lam[i][1]*lam[j][0]
```

6.4 CADABRA2

```
sudo apt install cadabra2
```

Python bindings available. Best for tensor operations in field theory.

6.5 FORM

Industrial-scale. No Python API subprocess only:

```
import subprocess
result = subprocess.run(['form', 'script.frm'], capture_output=True)
```

Part IV

Research Frontiers

Chapter 7

Research Frontiers

7.1 The Amplituhedron

Scattering amplitudes as volumes of positive geometries in $G(k, n)$.

7.2 Celestial Amplitudes

Amplitudes in Mellin space, 2D CFT structure, BMS symmetry.

7.3 Single-Minus Open Problems

From arXiv:2602.12176:

- Loop-level generalizations
- Soft factorization properties
- Relation to celestial amplitudes

7.4 Twistor String Theory

Degree-k curves in twistor space, CHY scattering equations.

Part V

Literature Review

Chapter 8

Literature Review and Paper Notes

This chapter contains our own notes, breakdowns, and computations related to important papers in the amplitudes literature.

8.1 A Brief Introduction to Modern Amplitude Methods

Paper: Lance J. Dixon, "A brief introduction to modern amplitude methods", arXiv:1310.5353 (2013).

This is an excellent pedagogical review based on lectures at the 2012 CERN Summer School and TASI 2013. It is one of the best entry points into modern amplitude methods.

8.1.1 Key Concepts Covered

- **Color decomposition**: Trace-based decomposition of QCD amplitudes into color-ordered partial amplitudes.
- **Spinor helicity formalism**: Review of massless spinors, angle and square brackets, polarization vectors expressed as spinor bilinears.
- **Factorization**: Universal soft and collinear factorization of tree-level gauge theory amplitudes.
- **Parke-Taylor MHV amplitudes**: The famous one-term formula for maximally helicity violating gluon scattering.
- **BCFW recursion**: On-shell recursion relations at tree level.
- **Unitarity methods**: Brief introduction to how these ideas extend to loop level.

8.1.2 Important Results & Equations

Parke-Taylor MHV formula (color-ordered):

$$A_n^{\text{tree}}(1^-, 2^-; 3^+, \ldots, n^+) = \frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle \cdots \langle n1\rangle} \tag{8.1}$$

BCFW recursion relation:

$$A_n = \sum_P A_L(z_P) \frac{1}{P^2(z_P)} A_R(z_P) \tag{8.2}$$

where the sum is over factorization channels and z_P is the value that puts the internal propagator on-shell.

Polarization vectors in spinor helicity:

$$\varepsilon_\mu^+(k;q) = \frac{\langle q|\gamma_\mu|k]}{\sqrt{2}\langle qk\rangle}, \tag{8.3}$$

$$\varepsilon_\mu^-(k;q) = \frac{[q|\gamma_\mu|k\rangle}{\sqrt{2}[qk]}. \tag{8.4}$$

8.1.3 Why This Paper Matters for Our Project

This paper connects directly to our existing work: - It builds on the spinor technology from Srednicki Ch. 48 and Ch. 60. - The Parke-Taylor formula is the seed amplitude we will verify with Cadabra2. - BCFW recursion is the next major topic after MHV (see Ch. 05 in our notes).

We will add detailed derivations, Cadabra2 checks of the spinor identities, and explicit computations of low-point MHV amplitudes in subsequent sections.

8.2 Single-Minus Gluon Tree Amplitudes Are Nonzero

Paper: Guevara, Lupsasca, Skinner, Strominger, Weil (on behalf of OpenAI), "Single-minus gluon tree amplitudes are nonzero", arXiv:2602.12176v2 (2025).

Significance: This is the first physics discovery made by an AI system (GPT-5.2 Pro). The key formula was *conjectured* by GPT and then *proved* by a new internal OpenAI model.

8.2.1 The Discovery

It was long believed that single-minus tree-level gluon amplitudes $\mathcal{A}_n(1^-, 2^+, \ldots, n^+)$ vanish at tree level for generic kinematics. The standard argument uses a clever choice of reference spinor to show all polarization vectors are orthogonal, so the amplitude requires more powers of momenta in the numerator than vertices can supply.

The **loophole**: when all angle brackets vanish ($\langle ij\rangle = 0$ for all i, j), the reference spinor choice becomes singular. This "half-collinear" regime exists in $(2, 2)$ Klein signature or for complexified momenta.

8.2.2 Key Equations

The stripped amplitude ansatz:

$$\mathcal{A}_n = i^{2-n} A_{1\cdots n} \prod_{a=2}^{n} \delta(\langle 1a\rangle) \cdot \delta^2\left(\sum_{i=1}^{n} \langle ri\rangle \tilde{\lambda}_i\right) \tag{8.5}$$

The main result (Eq. 16 of the paper):

$$A_{1\cdots n}\Big|_{\mathcal{R}_1} = \frac{1}{2^{n-2}} \prod_{m=2}^{n-1} \left(\mathrm{sg}_{m,m+1} + \mathrm{sg}_{1,2\cdots m}\right) \tag{8.6}$$

where $\mathrm{sg}_{ij} = \mathrm{sg}([\tilde{\lambda}_i \tilde{\lambda}_j])$ and $\mathcal{R}_1$ is the kinematic region where particle 1 has $\omega_1 < 0$ and all others have $\omega_a > 0$.

Concrete low-point examples:

$$A_{123}\big|_{\mathcal{R}_1} = \tfrac{1}{2}(\mathrm{sg}_{23} + \mathrm{sg}_{12}), \tag{8.7}$$

$$A_{1234}\big|_{\mathcal{R}_1} = \tfrac{1}{4}(\mathrm{sg}_{23} + \mathrm{sg}_{12})(\mathrm{sg}_{34} + \mathrm{sg}_{41}). \tag{8.8}$$

Each factor is $\frac{1}{2}(\pm 1 \pm 1) \in \{-1, 0, +1\}$, so the stripped amplitude is **piecewise-constant**.

8.2.3 Proof Strategy

The proof proceeds in three steps:

1. Show that in $\mathcal{R}_1$, the vertex function $V_{\tilde{\lambda}_2 \cdots \tilde{\lambda}_n} = 0$ (a causality condition).

2. Show the Berends–Giele recursion collapses: $A_{1\cdots n} = \bar{V}_{\tilde{\lambda}_2 \cdots \tilde{\lambda}_n}$.

3. Evaluate $\bar{V}$ using momentum conservation to recover Eq. (8.6).

A key ingredient is the "master identity" (App. A), a generalization of a standard distributional identity involving sign functions and δ-functions.

8.2.4 Open Directions (from the paper)

The authors note several extensions we should explore:

- General formula outside $\mathcal{R}_1$ (they give explicit results through $n = 6$)
- Graviton amplitude generalization
- Supersymmetric extensions
- Connection to $\mathcal{L}w_{1+\infty}$ algebra and celestial holography
- Mellin transforms yielding Lauricella functions
- Possible yet-simpler expressions via different variable choices

8.2.5 Why This Paper Matters for Our Project

This paper demonstrates that:

1. AI systems can make genuine discoveries in scattering amplitudes.
2. The discovery method was pattern recognition from computed data (GPT conjectured the formula from $n = 3, \ldots, 6$ examples).
3. The proof was also AI-assisted.

Our goal is to reproduce this methodology and find new results in related areas.